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MILITARY SPECIFICATION

# CLOTH, WIND RESISTANT SATEEN, COTTON AND NYLON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This document covers sateen cloth made of cotton and nylon blend.

1.2 Classification. The cloth shall be of the following classes as specified (see 6.2).

Class 1 - Dyed
Class 2 - Dyed and Quarpel treated
Class 3 - Woodland camouflage printed and Quarpel treated

## 2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-285 - Thread, Polyester
PPP-P-1134 - Packaging of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

2-1-1516 - Woodland Pattern - 48 inches
2-1-1516B - Woodland Pattern - 60 inches

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Copies may be obtained from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1424 - Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)



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(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard sample. The dyed, dyed and treated, or printed and treated finished cloth shall match the standard sample for shade and appearance and shall, unless otherwise indicated (see 3.4.5) be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2 and 6.7).

3.3 Materials (see 6.9).

3.3.1 Cotton. The cotton shall be carded and drawn.

3.3.2 Nylon. The nylon shall be first quality, high tenacity, semi-dull staple having a nominal cut length of 1-1/2 inches and a round cross section with a single nominal denier of 2.25 - 2.5.

3.3.2.1 Nylon staple. The nylon staple shall be carded and drawn.

3.3.3 Yarn. The yarns shall be singles and contain a blend of 50 ± 5 percent nylon and the remaining percentage cotton, based on the dry weight of the desized cloth prior to treatment. Testing shall be as specified in 4.4.3.

3.4 Color.

3.4.1 Classes 1 and 2. The color of the finished cloth shall be as specified and shall match the standard sample.

* 3.4.1.1 Spectral reflectance, class 2. When shade Olive Green 107 is specified, the spectral reflectance values of the finished cloth shall meet the requirements specified in table I when tested as specified in 4.4.3.



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TABLE I. Spectral reflectance (percent) limits for class 2 when shade OG 107 is specified

| Wavelength, nanometers | Reflectance Min. | Max. | Wavelength nanometers | Reflectance Min. | Max. |
|---|---|---|---|---|---|
| 600 | -- | 9 | 740 | 13 | -- |
| 620 | -- | 9 | 760 | 20 | -- |
| 640 | -- | 9 | 780 | 27 | -- |
| 660 | -- | 9 | 800 | 35 | -- |
| 680 | -- | 10 | 820 | 42 | -- |
| 700 | -- | 13 | 840 | 50 | -- |
| 720 | 8 | -- | 860 | 50 | -- |

3.4.2 Class 3. The cloth shall be dyed to Ground Shade approximating Light Green 354. The Woodland camouflage pattern shall be obtained by roller or screen printing using four rollers or screens, as appropriate, for the Light green 354, Dark Green 355, Brown 356 and Black 357 areas of the pattern (see 6.4.2).

3.4.2.1 Pattern execution, class 3. The pattern shall reproduce the standard sample in respect to design, colors and registration of the respective areas. The pattern repeat of the dyed, printed and finished cloth shall be 27.25 + 1.25 - 2.50 inches in the warp direction. Each pattern area shall show solid coverage; skitteriness exceeding that shown by the standard sample is not referenced for pattern execution or design, a pattern drawing shall be provided and the pattern on the cloth shall match the drawing (see 2.1, 6.2 and 6.3).

3.4.2.2 Spectral reflectance, class 3. The spectral reflectance of the colors in the camouflage printed finished cloth shall conform to the requirements specified in table II when tested as specified in 4.4.3.

*TABLE II. Spectral reflectance requirements, class 3

| Wavelengths nanometers | Reflectance Values (percent) Black 357 Min. | Max. | Light Green 354 Min. | Max. | Dark Green 355 and Brown 356 Min. | Max. |
|---|---|---|---|---|---|---|
| 600 | -- | 10 | 8 | 20 | 3 | 10 |
| 620 | -- | 10 | 8 | 20 | 3 | 10 |
| 640 | -- | 10 | 8 | 20 | 3 | 10 |
| 660 | -- | 10 | 8 | 22 | 3 | 12 |
| 680 | -- | 10 | 10 | 27 | 3 | 16 |
| 700 | -- | 10 | 18 | 40 | 5 | 16 |
| 720 | -- | 10 | 22 | 53 | 7 | 20 |



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*TABLE II. Spectral reflectance requirements, class 3 (cont'd)

| Wavelengths nanometers | Reflectance Values (percent) | | | | | |
|---|---|---|---|---|---|---|
| | Black 357 | | Light Green 354 | | Dark Green 355 and Brown 356 | |
| | Min. | Max. | Min. | Max. | Min. | Max. |
| 740 | -- | 10 | 30 | 64 | 12 | 28 |
| 760 | -- | 10 | 35 | 73 | 18 | 36 |
| 780 | -- | 10 | 40 | 80 | 26 | 44 |
| 800 | -- | 10 | 45 | 85 | 34 | 52 |
| 820 | -- | 10 | 50 | 88 | 42 | 60 |
| 840 | -- | 10 | 55 | 90 | 50 | 68 |
| 860 | -- | 10 | 60 | 91 | 56 | 74 |

3.4.3 Labile sulfur. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid shall be chosen and applied so that the dyed cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4.3. When a standard sample is not available, the dyed cloth shall show no more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.4.3.

* 3.4.4 Matching. The color and appearance of the dyed, dyed and treated, or printed and treated finished cloth shall match the standard sample when viewed under filtered tungsten lamps which approximate artificial daylight having a correlated color temperature of 7000 ± 500 K with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incancescent lamplight at 2300 ± 100 K.

3.4.5 Colorfastness.

* 3.4.5.1 Classes 1 and 2. The dyed or dyed and treated finished cloth shall show fastness to laundering (after 3 cycles), perspiration, light, and crocking equal to or better than the standard sample or equal to or better than the rating of "good" when tested as specified in 4.4.3.

* 3.4.5.2 Class 3. The camouflage printed and water repellent treated cloth shall show fastness to laundering (after 3 cycles), perspiration, light, and crocking equal to or better than the standard sample or equal to or better than a rating of "good" for each of the pattern areas, except Black 357 which shall be equal to or better than a rating of "fair" when tested as specified in 4.4.3. The cloth shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating for crocking not less than 3.5 for all the pattern areas, except Black 357 which shall have an AATCC Chromatic Transference Scale rating of not lower than 1.5, when tested as specified in 4.4.3.



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3.5 Physical requirements. The physical requirements of the dyed, dyed and treated, or printed and treated finished cloth shall be as specified in table III when tested as specified in 4.4.3.

TABLE III. Physical requirements

| Weight oz/sq. yd. Minimum | Yarns per inch Minimum | | Breaking strength pounds Minimum | | Tearing strength pounds Minimum | | Weave 1/ |
|---|---|---|---|---|---|---|---|
| | Warp | Filling | Warp | Filling | Warp | Filling | |
| 8.5 | 126 | 72 | 225 | 225 | 9.0 | 9.0 | 5-harness sateen |

1/ The weave when viewed from the face side (see 3.6) shall appear as shown on figure 1.

* 3.5.1 Width. The width of the cloth shall be as specified (see 6.2), and shall be the minimum acceptable width, inclusive of selvages when fly shuttle looms or shuttleless with tuck-in selvage looms are used. For all other shuttleless looms the width measurement shall be made between the last warp yarn on each side excluding the protruding fringe(s).

3.6 Finish. The cloth shall be dyed, or dyed and printed on the filling effect side as the face. The cloth shall be closely singed, desized, mercerized and dyed or dyed and printed. The class 2 and 3 cloths shall be given a water repellent treatment as specified in 3.10.

3.6.1 Nonfibrous material. The starch and protein content including chloroform-soluble and water-soluble material and prior to the water repellent treatment shall not exceed 2.0 percent when tested as specified 4.4.3.

3.7 pH. The pH value of the water extract of the finished cloth shall be no less than 5.0 and no more than 8.5 when tested as specified in 4.4.3.

3.8 Dimensional stability. The shrinkage or elongation both in the warp and in the filling of the finished cloth shall not be greater than 2.5 percent for the individual sample unit and not greater than 2.0 percent for the lot average when tested as specified in 4.4.3. The preshrinakge process used shall not be identified by name or trademark either on the cloth, ticket or package.



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3.9 Face identification (classes 1 and 2). The filling effect side of the class 1 and 2 cloths shall be identified as the face side by stamping that side with the word "Face" at each end of the roll.

3.10 Water repellency and air permeability (classes 2 and 3). The cloth shall be given a approved Quarpel type (see 6.6) water-repellent treatment and shall conform to the water-repellency and air-permeability requirements of table IV when tested as specified in 4.4.3. The use of materials other than approved water-repellents and sodium acetate buffer (and acetic acid) is prohibited.

TABLE IV. Water repellency and air permeability requirements

| | Hydrostatic pressure (centimeters) | Dynamic absorption (percent) | Air permeability cu. ft./min/sq/ ft. |
|---|---|---|---|
| | Minimum Average | Maximum Average | Maximum Average |
| Initial | 30 1/ | 20 2/ | - |
| After 3 launderings | 30 1/ | - | 7.0 3/ |
| After 15 launderings | - | 20 2/ | - |

1/ No individual specimen shall fall below 25 centimeters.

2/ Results shall be basesd on the average of all the sample units from the lot. No sample unit (average of 2 specimens) shall exceed 25 percent.

3/ No sample unit (average of 5 specimens) shall exceed 7.5.

3.11 Resistance to organic liquid (classes 2 and 3). The finished cloth shall show no wetting by n-tetradecane initially and after 15 launderings when tested as specified 4.4.3.

3.12 Spray rating (classes 2 and 3). The measurements of the three individual determinations on the sample unit for spray rating shall be equal to or better than 90, 90, 80 when tested as specified 4.4.3.

3.13 Seam efficiency. The dyed, dyed and treated, or printed and treated finished cloth shall have a seam efficiency of not less than 80 percent when tested as specified in 4.4.3.

3.14 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 50 yards. Each length shall be put up in full width rolls as specified in 5.1.



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3.15 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.16 Workmanship. The finished cloth shall conform to the quality of product established by this document. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contactor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for appearance, color, and finish defects and tested for the characteristics specified in table V. The presence of any defect or failure of any test shall be cause for rejection of the first article.



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4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 End item examination.

4.4.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in section I of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.4.2.1.1, except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scales (see 6.8), "F" or "2" as applicable for slubs and "D" for knots shall be scored. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 30 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 45 points. If one roll exceeds 45 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality. The lot shall be unacceptable if one or more rolls in the second sample exceeds 45.0 points per 100 square yards. Points computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.4.2.1.1 Demerit points. Demerit points shall be assigned as follows:

| | | |
|---|---|---|
| For defects up to 3 inches in any direction | - | one point |
| For defects exceeding 3 inches, but not exceeding 6 inches in any dimension | - | two points |
| For defects exceeding 6 inches, but not exceeding 9 inches in any dimension | - | three points |
| For defects exceeding 9 inches in any dimension | - | four points |

The following defects, when present, shall be scored four points for each yard in which they occur:



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Objectionable odor
Baggy, ridgy, or wavy cloth
Width less than specified
Edge ravels when pulled outward
Slack or tight selvages 1/
Overall uncleanness
Excessive neppiness
Poor dye penetration, mottled, streaky, or cloudy
Pattern design not equal to standard sdample (class 3)
Incorrect color in any part of the pattern (class 3)
Pattern repeat less than 24.75 inches or more than 28.50 inches (class 3)
Skitteriness (mottled, uneven color) of pattern exceeds that is shown by the standard sample (class 3)
Excessive feathering or spew (fuzziness at color boundries) of pattern as compared to the sample (class 3)
Excessive grinning (off register, gap where ground shade shows through) of pattern as compared to the standard sample (class 3)
Excessive haloing or trapping (overlapping of colors) of pattern as compared to the standard sample (class 3)
Overall application of water repellent not uniform (classes 2 and 3)
Tackiness (sticky to touch) (classes 2 and 3)
Design not printed on face side of cloth (class 3)

1/ To determine the presence of unacceptable selvage conditions, the following procedure shall be observed: During the visual examination, the perch shall be stopped a minimum of three times for each roll in the sample, the tension removed and the finished cloth examined for the selvage conditions. Suspect rolls should be removed from the perch and unrolled on the floor or a table and further examined for the condition. A waviness in the selvage or significant ripples diagonally across the width of the fabric is an indication of slack or tight selvages.

4.4.2.2 Length examination.

4.4.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.



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4.4.2.3 Shade and finish appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade and finish appearance. Any roll in the sample off shade (compared to the standard sample and applicable shade range) or shaded side to side, side to center, or end to end, or any roll that does not have the same finish appearance as the standard sample shall be cause for rejection of the entire lot represented by the sample.

4.4.2.4 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample:

Face identification missing from either or both ends (classes 1 and 2)
Face identification on wrong side (classes 1 and 2)
Preshrinkage process identified by name or trademark on cloth or ticket
Not labeled in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act

4.4.3 End item testing. The cloth shall be tested for the characteristics listed in table V. The methods of testing specified in FED-STD-191 whenever applicable and as listed in table V shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be 1/4 yard full width of the dyed cloth (prior to treatment) for determination of fiber content and non-fibrous material content and 5 continuous yards full width for all other physical and chemical tests. The lot shall be unacceptable if one or more sample units or the lot average for dynamic absorption, air permeability or dimensional stability fail to meet any requirement specified. The sample size shall be in accordance with the following:

| Lot size (yards) | Sample size (sample units) |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and over | 5 |

TABLE V. End item tests

| Characteristic | Requirement paragraph | Test method |
|---|---|---|
| Material: | | |
| Cotton fiber identification | 3.3.1 | 1200 1/ |
| Nylon: | | |
| Identification | 3.3.2 | 1530 1/ |
| Luster | 3.3.2 | 1/ |
| Denier | 3.3.2 | 1/ |



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TABLE V. End item tests

| Characteristic | Requirement paragraph | Test method |
|---|---|---|
| Fiber content: | | |
| Cotton content | 3.3.3 | 2530 1/ 2/ |
| Nylon content | 3.3.3 | 2530 1/ |
| Spectral reflectance (classes 2 and 3) | 3.4.1.1 and 3.4.2.2 | 4.5.1 |
| Presence of labile sulfur | 3.4.3 | 2020 1/ |
| Colorfastness to: | | |
| Laundering (after 3 cycles) | 3.4.5 | 5610 3/ 4/ |
| Perspiration | 3.4.5 | 5680 |
| Light | 3.4.5 | 5660 |
| Crocking | 3.4.5 | 5651 |
| Weight | 3.5 | 5041 |
| Yarns per inch | 3.5 | 5050 |
| Breaking strength | 3.5 | 5100 |
| Tearing strength | 3.5 | ASTM D 1424 |
| Weave | 3.5 | Visual 5/ |
| Singed | 3.6 | 1/ |
| Desized | 3.6 | 1/ |
| Mercerized | 3.6 | 1/ |
| Nonfibrous material content | 3.6.1 | 2611 |
| pH | 3.7 | 2811 |
| Shrinkage in laundering (dimensional stability) | 3.8 | 5550 |
| Hydrostatic pressure (classes 2 and 3): | | |
| Initial | 3.10 | 5514 |
| After laundering | 3.10 | 5556, 5514 |
| Dynamic absorption (classes 2 and 3): | | |
| Initial | 3.10 | 5500 |
| After laundering | 3.10 | 5556, 5500 |
| Air permeability (after laundering) (classes 2 and 3) | 3.10 | 5556, 5450 |
| Water repellents (classes 2 and 3) | 3.10 | 6/ |
| Resistance to organic liquid (classes 2 and 3): | | |
| Initial | 3.11 | 4.5.2 |
| After 15 launderings (cotton method) | 3.11 | 4.5.2 7/ |
| Water repellency (classes 2 and 3): | | |
| Spray rating | 3.12 | 5526 |
| Seam efficiency | 3.13 | 5110 8/ |

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.



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2/ The cotton content shall be calculated as follows:

$$\text{Cotton content, percent} = \frac{R}{S} \times 100$$

R = weight of residual fibers, and
S = weight of dry, desized, specimen

3/ The specimens must be dried after each of the 3 laundering cycles.

4/ On the color transfer cloth evaluation, only the stain on the cotton and nylon fibers of the color transfer cloth shall be evaluated.

5/ One determination per sample unit and result reported as "pass" or "fail".

6/ The contractor shall report the water repellents used, and certify that no other material (except the specified buffer and acetic acid) has been added.

7/ Specimens shall be subjected to 15 complete cycles (wash and dry) prior to determinations of resistance to organic liquid after laundering.

8/ The needle shall measure 0.049 ± 0.001 inch across the blade of the eye. The thread shall be polyester thread conforming to type I, class 1, sub-class A of V-T-285 for class 1 cloth, and polyester thread conforming to type I, class 1, sub-class B of V-T-285 for classes 2 and 3 cloth, with size F for the needle and size E for the looper.

4.4.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1134.

4.5 Methods of inspection.

4.5.1 Spectral reflectance test. Reflectance data shall be obtained from 600 to 860 nanometers (nm) for each color relative to barium sulfate, the preferred white reference. Other reference white materials may be used (Halon, magnesium oxide, or vitrolite tile), provided they are calibrated to absolute white (see 6.10). The spectral band width at 860 nm shall be less than 25 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer (see 6.5) shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates in the visible spectrum either CIE Source A or CIE Source D65. Specimens of Black 357 shall be measured as a single layer backed with two layers of the same shade cut from the standard. Each of the remaining three colors in the Woodland pattern shall be measured as a single layer backed with three layers of its same color cut from the standard. Readings will be taken on a minimum of two different areas, and the data averaged. The specimen shall be viewed at an angle no greater than 10 degrees from normal. Photometric accuracy of the spectrophotometer shall be within 1 percent and the wavelength accuracy shall be within 2 nm. The standard aperture



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size used in the color measurement device shall be 1.0 to 1.25 inches in diameter. When the measured reflectance values for any color at four or more of the listed wavelengths do not meet the limits specified in table II, it shall be considered a test failure.

4.5.2 Resistance to organic liquid test. Place a small specimen of the cloth on a smooth horizontal surface, face up. Using a pipette or eye dropper, gently deposit one drop of n-tetradecane on the surface of the specimen. After 1 minute examine the specimen under light at an angle. Absence of light reflectance at the cloth-drop interface shall be taken as evidence of wetting. Three specimens (or areas) taken at various locations across the sample unit shall be tested. Evidence of wetting on one or more specimens shall be considered a test failure.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial. The cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Levels A, B and Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth is intended for use in clothing where a high degree of wind resistance, water resistance and thermal resistance is of prime importance.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Class required (see 1.2).
c. When a first article is required (see 3.2, 4.3 and 6.7).
d. Color of cloth required (see 3.4).
e. Woodland camouflage pattern drawing if required (see 3.4.2.1).
f. Width of cloth required (see 3.5.1).
g. Length of roll, if other than specified (see 3.14).
h. Selection of applicable levels of put-up, preservation and packing (see 5.1 and 5.2).


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6.3 Standard sample and pattern drawing. For access to the standard shade sample (see 3.1) and the Woodland camouflage pattern drawing if applicable (see 3.4.2.1) address the contracting activity issuing the invitation for bids.

6.4 Dyeing.

6.4.1 Olive Green 107. Olive Green 107 has been satisfactorily dyed with the following colorants:

| | | |
|---|---|---|
| Nylon | Acid Yellow | 219 |
| | Acid Blue | 258 |
| | Acid Orange | 162 |
| Cotton | Vat Brown | 57 |
| | Vat Brown | 3 |
| | Vat Green | 1 |
| | Vat Yellow | 2 |
| | Vat Orange | 2 |
| | Cibanone Olive Green NY | |

The above formulations are suggested, and are not mandatory.

6.4.2 Woodland camouflage printing, class 3. The areas of the pattern for class 3 have been found to be satisfactorily when dyed or printed with the following colorants. The dyeing of the ground shade approximating Light Green 354 was accomplished using the acid and vat dyes listed below.

Light Green 354

Vat Orange 2
Vat Green 1
Vat Yellow 2

Dark Green 355

Vat Brown 3
Vat Green 1
Vat Yellow 2

Brown 356

Vat Brown 57
Bat Brown 3
Vat Yellow 2
Vat Green 1

Black 357

Sulfur Black 6
Vat Blue 20
Vat Brown 3 Vat Black 11

Ground Shade

Acid Blue 258
Tectilon Orange 4R
Vat Orange 2
Vat Green 1
Vat Yellow 2



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Satisfactory printing of shades Light Green 354, Dark Green 355, and Brown 356 has been accomplished by the use of vat dyes only. Shade Black 357 has been satisfactorily printed with a combination of vat and sulfur dyes.

6.5 Spectrophotometers. The basic instrument for measuring the infrared reflectance is the General Electric Diano Hardy spectrophotmeter, but other instruments such as the Beckman DU of DK Diano Match-Scan, the Cary 14 or the Hunter D54P-IR spectrophotometers can be used (see 4.5.1).

6.6 Quarpel water repellent. Approval of components and combination is the responsibility of the U.S. Army Natick Research, Development and Engineering Center, Natick, MA, and is based on more extensive tests, including those for toxicity, which are not set forth in this document. Because of the time necessary to conduct full evaluation, only those chemical treatments already approved and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement.

* 6.7 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.8 Fabric defect scales. Sears Fabric Defect Replica Kits are available from:

Sears, Roebuck and Company
Department 817, (ATTN: BSC 23-29)
Sears Tower
Chicago, IL 60684

6.9 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

* 6.10 Source of material. Barium sulphate and Halon of suitable quality for use as a white standard is available from Eastman Kodak Company. Suitable white tiles are available from the National Bureau of Standards or spectrophotometer manufacturers (see 4.5.1).

6.11 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



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Custodians:

Army - GL
Navy - NU
Air Force - 99

Review activities:

Army - ER, MD
DLA - CT

User activities:

Navy - MC
Air Force - 82

Preparing activity:

Army - GL

Project No. 8305-0054



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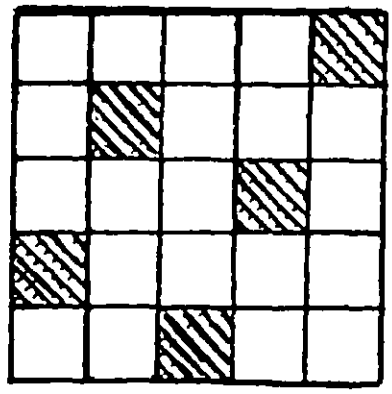


Figure 1



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-43191F | Cloth, Wind Resistant Sateen, Cotton and Nylon |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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